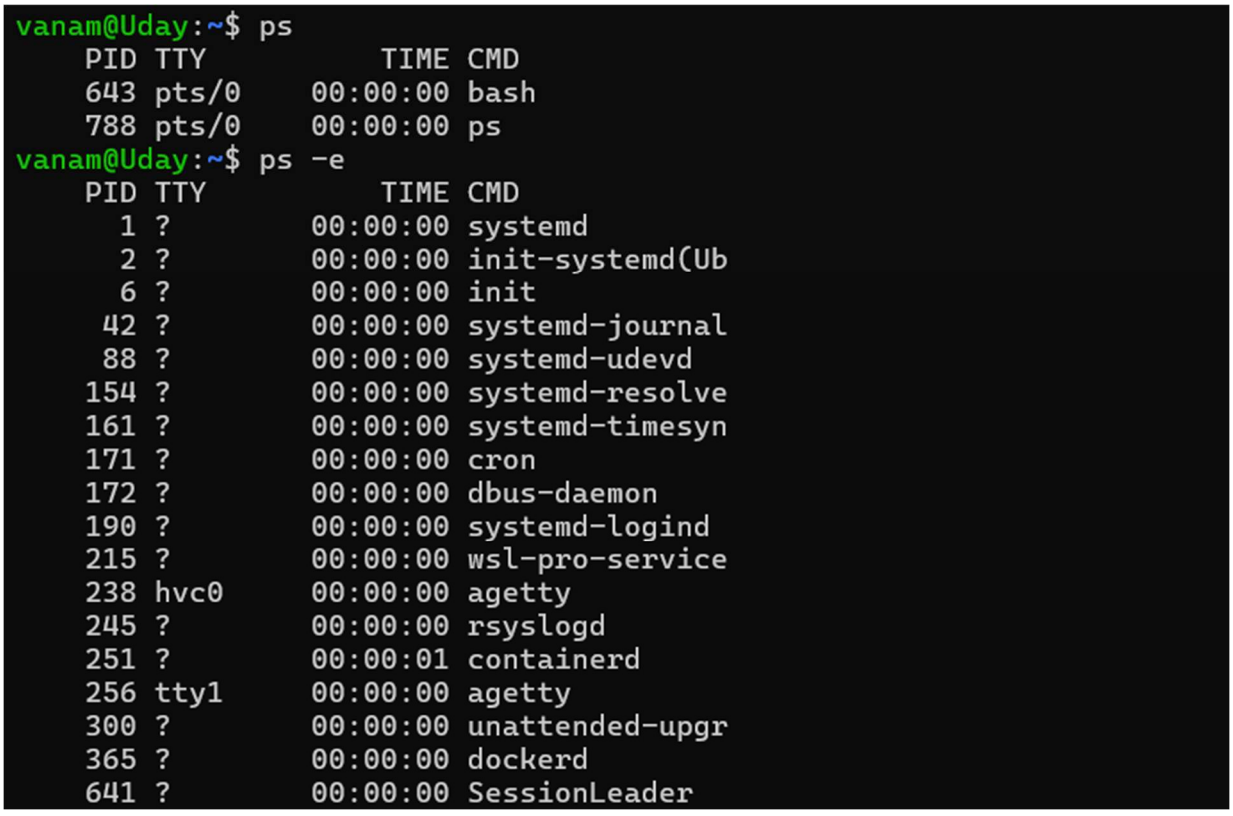
Process Management:

PS

Option	Meaning
-e	Select all processes
-f	Full-format listing
a	Show processes associated with terminals from all users
u	User-oriented format
x	Include processes without a controlling terminal



```
vanam@Uday:~$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root           1       0  0 08:54 ?        00:00:00 /sbin/init
root           2       1  0 08:54 ?        00:00:00 /init
root           6       2  0 08:54 ?        00:00:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --
root          42       1  0 08:54 ?        00:00:00 /usr/lib/systemd/systemd-journald
root          88       1  0 08:54 ?        00:00:00 /usr/lib/systemd/systemd-udevd
systemd+     154       1  0 08:54 ?        00:00:00 /usr/lib/systemd/systemd-resolved
systemd+     161       1  0 08:54 ?        00:00:00 /usr/lib/systemd/systemd-timesyncd
root         171       1  0 08:54 ?        00:00:00 /usr/sbin/cron -f -P
message+     172       1  0 08:54 ?        00:00:00 @dbus-daemon --system --address=systemd: --nofork --nopidfile --syst
root         190       1  0 08:54 ?        00:00:00 /usr/lib/systemd/systemd-logind
root         215       1  0 08:54 ?        00:00:00 /usr/libexec/wsl-pro-service
root         238       1  0 08:54 hvc0     00:00:00 /sbin/agetty -o -p -- \u --noclear --keep-baud - 115200,38400,9600 v
syslog       245       1  0 08:54 ?        00:00:00 /usr/sbin/rsyslogd -n -iNONE
root         251       1  0 08:54 ?        00:00:01 /usr/bin/containerd
root         256       1  0 08:54 tty1     00:00:00 /sbin/agetty -o -p -- \u --noclear - linux
root         300       1  0 08:54 ?        00:00:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-s
root         365       1  0 08:54 ?        00:00:00 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.so
root         641       2  0 08:54 ?        00:00:00 /init
root         642     641  0 08:54 ?        00:00:00 /init
vanam        643     642  0 08:54 pts/0    00:00:00 -bash
root         644       2  0 08:54 pts/1    00:00:00 /bin/login -f
```

```
vanam@Uday:~$ ps aux
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.2  0.1  21752 12228 ?        Ss   08:54   0:00 /sbin/init
root           2  0.0  0.0   3072  1664 ?        Sl   08:54   0:00 /init
root           6  0.0  0.0   3072  1792 ?        Sl   08:54   0:00 plan9 --control-socket 7 --log-level 4 --server-fd 8
root          42  0.0  0.1  50440 15488 ?        S<s  08:54   0:00 /usr/lib/systemd/systemd-journald
root          88  0.0  0.0  25152  6016 ?        Ss   08:54   0:00 /usr/lib/systemd/systemd-udevd
systemd+     154  0.0  0.1  21344 12800 ?        Ss   08:54   0:00 /usr/lib/systemd/systemd-resolved
systemd+     161  0.0  0.0  91036  7552 ?        Ssl  08:54   0:00 /usr/lib/systemd/systemd-timesyncd
root         171  0.0  0.0   4240  2432 ?        Ss   08:54   0:00 /usr/sbin/cron -f -P
message+     172  0.0  0.0   9604  4864 ?        Ss   08:54   0:00 @dbus-daemon --system --address=systemd: --nofork --n
root         190  0.0  0.1  17976  8064 ?        Ss   08:54   0:00 /usr/lib/systemd/systemd-logind
root         215  0.0  0.1 1830360 12544 ?       Ssl  08:54   0:00 /usr/libexec/wsl-pro-service
root         238  0.0  0.0   3164  1920 hvc0     Ss+  08:54   0:00 /sbin/agetty -o -p -- \u --noclear --keep-baud - 1152
syslog       245  0.0  0.0 222516  5504 ?        Ssl  08:54   0:00 /usr/sbin/rsyslogd -n -iNONE
root         251  0.2  0.5 2384476 46336 ?       Ssl  08:54   0:01 /usr/bin/containerd
root         256  0.0  0.0   3120  1792 tty1     Ss+  08:54   0:00 /sbin/agetty -o -p -- \u --noclear - linux
root         300  0.0  0.2 107044 21632 ?        Ssl  08:54   0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unatt
root         365  0.2  1.0 2723408 81136 ?       Ssl  08:54   0:00 /usr/bin/dockerd -H fd:// --containerd=/run/container
root         641  0.0  0.0   3080   896 ?        Ss   08:54   0:00 /init
root         642  0.0  0.0   3096  1028 ?        S    08:54   0:00 /init
vanam        643  0.0  0.0   6076  5120 pts/0    Ss   08:54   0:00 -bash
root         644  0.0  0.0   6692  4352 pts/1    Ss   08:54   0:00 /bin/login -f
vanam        691  0.0  0.1  20328 11264 ?        Ss   08:54   0:00 /usr/lib/systemd/systemd --user
vanam        692  0.0  0.0  21160  3512 ?        S    08:54   0:00 (sd-pam)
```

```
vanam@Uday:~$ ps aux | grep python
root         300  0.0  0.2 107044 21632 ?        Ssl  08:54   0:00 /usr/bin/python3 /usr/share/unattended
ended-upgrade-shutdown --wait-for-signal
vanam        793  0.0  0.0   4096  1920 pts/0    S+   09:03   0:00 grep --color=auto python
vanam@Uday:~$
```

Top

Key Action

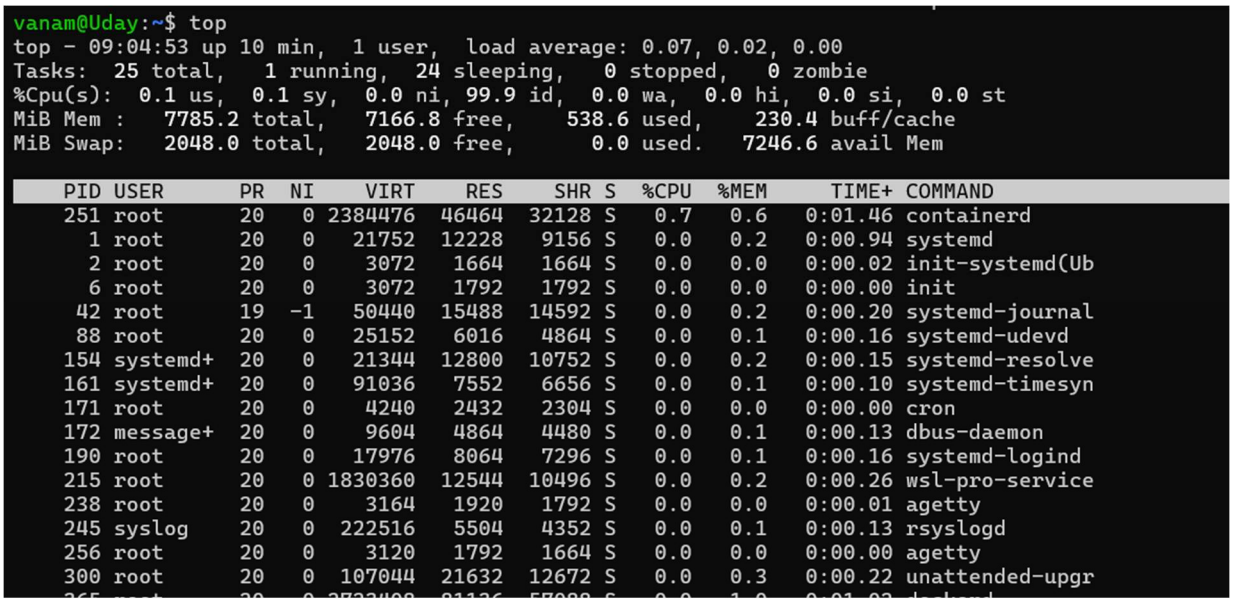
P Sort by CPU usage

M Sort by memory usage

1 Toggle individual CPU-core statistics

k Request to terminate a process by PID

q Quit



P

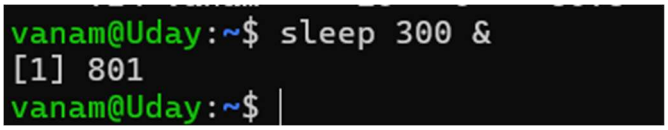
```
  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
  251 root      20   0 2384476  46464  32128 S   0.3   0.6   0:01.50 containerd
    1 root      20   0   21752  12228   9156 S   0.0   0.2   0:00.94 systemd
    2 root      20   0    3072   1664   1664 S   0.0   0.0   0:00.02 init-systemd(Ub
    6 root      20   0    3072   1792   1792 S   0.0   0.0   0:00.00 init
   42 root      19  -1   50440  15488  14592 S   0.0   0.2   0:00.20 systemd-journal
   88 root      20   0   25152   6016   4864 S   0.0   0.1   0:00.16 systemd-udevd
  154 systemd+  20   0   21344  12800  10752 S   0.0   0.2   0:00.15 systemd-resolve
  161 systemd+  20   0   91036   7552   6656 S   0.0   0.1   0:00.10 systemd-timesyn
  171 root      20   0    4240   2432   2304 S   0.0   0.0   0:00.00 cron
  172 message+  20   0    9604   4864   4480 S   0.0   0.1   0:00.13 dbus-daemon
  190 root      20   0   17976   8064   7296 S   0.0   0.1   0:00.16 systemd-logind
  215 root      20   0 1830360  12544  10496 S   0.0   0.2   0:00.26 wsl-pro-service
  238 root      20   0    3164   1920   1792 S   0.0   0.0   0:00.01 agetty
  245 syslog    20   0  222516   5504   4352 S   0.0   0.1   0:00.13 rsyslogd
  256 root      20   0    3120   1792   1664 S   0.0   0.0   0:00.00 agetty
  300 root      20   0  107044  21632  12672 S   0.0   0.3   0:00.22 unattended-upgr
  365 root      20   0 2723408  81136  57088 S   0.0   1.0   0:01.02 dockerd
  641 root      20   0    3080    896    896 S   0.0   0.0   0:00.00 SessionLeader
  642 root      20   0    3096   1028    896 S   0.0   0.0   0:00.02 Relay(643)
  643 vanam     20   0    6208   5248   3584 S   0.0   0.1   0:00.09 bash
  644 root      20   0    6692   4352   3712 S   0.0   0.1   0:00.03 login
  691 vanam     20   0   20328  11264   9216 S   0.0   0.1   0:00.13 systemd
  692 vanam     20   0   21160   3512   1792 S   0.0   0.0   0:00.00 (sd-pam)
```

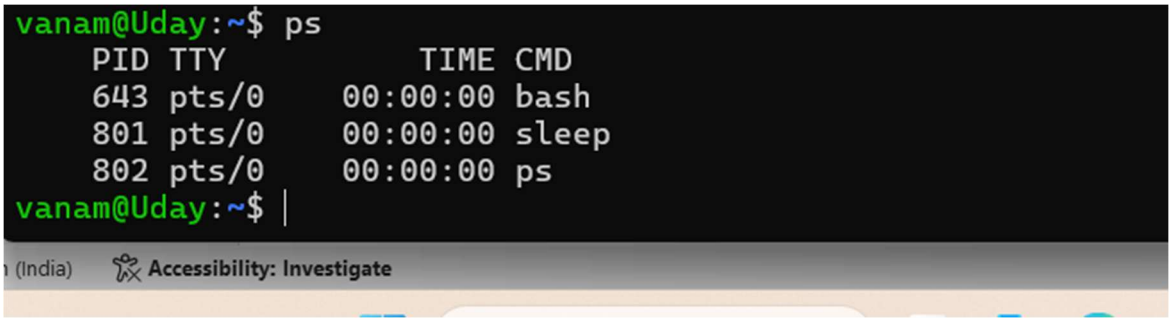


M

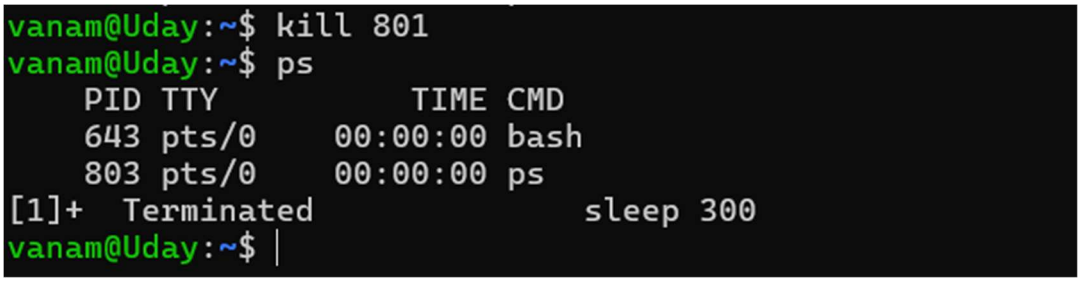
```
  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
    1 root      20   0   21752  12228   9156 S   0.0   0.2   0:00.94 systemd
    2 root      20   0    3072   1664   1664 S   0.0   0.0   0:00.02 init-systemd(Ub
    6 root      20   0    3072   1792   1792 S   0.0   0.0   0:00.00 init
   42 root      19  -1   50440  15488  14592 S   0.0   0.2   0:00.21 systemd-journal
   88 root      20   0   25152   6016   4864 S   0.0   0.1   0:00.16 systemd-udevd
  154 systemd+  20   0   21344  12800  10752 S   0.0   0.2   0:00.15 systemd-resolve
  161 systemd+  20   0   91036   7552   6656 S   0.0   0.1   0:00.10 systemd-timesyn
  171 root      20   0    4240   2432   2304 S   0.0   0.0   0:00.00 cron
  172 message+  20   0    9604   4864   4480 S   0.0   0.1   0:00.13 dbus-daemon
  190 root      20   0   17976   8064   7296 S   0.0   0.1   0:00.16 systemd-logind
  238 root      20   0    3164   1920   1792 S   0.0   0.0   0:00.01 agetty
  245 syslog    20   0  222516   5504   4352 S   0.0   0.1   0:00.13 rsyslogd
  251 root      20   0 2384476  46464  32128 S   0.0   0.6   0:01.51 containerd
  256 root      20   0    3120   1792   1664 S   0.0   0.0   0:00.00 agetty
  300 root      20   0  107044  21632  12672 S   0.0   0.3   0:00.22 unattended-upgr
  365 root      20   0 2723408  81136  57088 S   0.0   1.0   0:01.02 dockerd
  641 root      20   0    3080    896    896 S   0.0   0.0   0:00.00 SessionLeader
  642 root      20   0    3096   1028    896 S   0.0   0.0   0:00.02 Relay(643)
  643 vanam     20   0    6208   5248   3584 S   0.0   0.1   0:00.09 bash
  644 root      20   0    6692   4352   3712 S   0.0   0.1   0:00.03 login
  691 vanam     20   0   20328  11264   9216 S   0.0   0.1   0:00.13 systemd
  692 vanam     20   0   21160   3512   1792 S   0.0   0.0   0:00.00 (sd-pam)
```

Creating a Process





Killing a Process



df — Check filesystem disk space

Option	Meaning
-h	Human-readable sizes, such as MB and GB
-T	Display filesystem type
-i	Display inode usage instead of block usage

```
vanam@Uday:~$ df
Filesystem      1K-blocks      Used Available Use% Mounted on
none              3986020         0   3986020   0% /usr/lib/modules/6.6.87.2-microso
none              3986020         4   3986016   1% /mnt/wsl
drivers         478884860 427947196  50937664  90% /usr/lib/wsl/drivers
/dev/sdd       1055762868   7775864 994283532   1% /
none              3986020        80   3985940   1% /mnt/wslg
none              3986020         0   3986020   0% /usr/lib/wsl/lib
rootfs            3981004      2672   3978332   1% /init
none              3986020       552   3985468   1% /run
none              3986020         0   3986020   0% /run/lock
none              3986020         0   3986020   0% /run/shm
none              3986020        76   3985944   1% /mnt/wslg/versions.txt
```

H

```
vanam@Uday:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
none            3.9G     0  3.9G   0% /usr/lib/modules/6.6.87.2-microsoft-standard-WSL2
none            3.9G  4.0K  3.9G   1% /mnt/wsl
drivers         457G  409G   49G  90% /usr/lib/wsl/drivers
/dev/sdd       1007G  7.5G  949G   1% /
none            3.9G   80K  3.9G   1% /mnt/wslg
none            3.9G     0  3.9G   0% /usr/lib/wsl/lib
rootfs          3.8G  2.7M  3.8G   1% /init
none            3.9G  552K  3.9G   1% /run
none            3.9G     0  3.9G   0% /run/lock
none            3.9G     0  3.9G   0% /run/shm
none            3.9G   76K  3.9G   1% /mnt/wslg/versions.txt
none            3.9G   76K  3.9G   1% /mnt/wslg/doc
C:\             457G  409G   49G  90% /mnt/c
tmpfs           779M   20K  779M   1% /run/user/1000
```

T

```
vanam@Uday:~$ df -T
Filesystem     Type     1K-blocks      Used Available Use% Mounted on
none           overlay    3986020         0   3986020   0% /usr/lib/modules/6.6.87.2-microsoft-standard-WSL2
none           tmpfs      3986020         4   3986016   1% /mnt/wsl
drivers        9p       478884860 427947984  50936876  90% /usr/lib/wsl/drivers
/dev/sdd       ext4    1055762868   7775864 994283532   1% /
none           tmpfs      3986020        80   3985940   1% /mnt/wslg
none           overlay    3986020         0   3986020   0% /usr/lib/wsl/lib
rootfs         rootfs     3981004      2672   3978332   1% /init
none           tmpfs      3986020       552   3985468   1% /run
none           tmpfs      3986020         0   3986020   0% /run/lock
none           tmpfs      3986020         0   3986020   0% /run/shm
none           overlay    3986020        76   3985944   1% /mnt/wslg/versions.txt
none           overlay    3986020        76   3985944   1% /mnt/wslg/doc
C:\            9p       478884860 427947984  50936876  90% /mnt/c
```

du — Check directory and file sizes

```
vanam@Uday:~$ du file1.txt
4       file1.txt
vanam@Uday:~$
```

```
vanam@Uday:~$ du -h file1.txt
4.0K    file1.txt
```

free — Check memory usage

Option	Meaning
-h	Human-readable units
-m	Display values in MiB
-g	Display values in GiB
-s 2	Repeat the display every 2 seconds

```
vanam@Uday:~$ free
               total        used        free      shared  buff/cache   available
Mem:         7972040      549720     7324924        3572      251768     7422320
Swap:        2097152           0     2097152
```

H

```
vanam@Uday:~$ free -h
               total        used        free      shared  buff/cache   available
Mem:           7.6Gi       537Mi       7.0Gi       3.5Mi       245Mi       7.1Gi
Swap:          2.0Gi          0B       2.0Gi
```

M

In mb

```
vanam@Uday:~$ free -m
               total        used        free      shared  buff/cache   available
Mem:            7785         537        7152           3         245        7247
Swap:           2048           0        2048
```

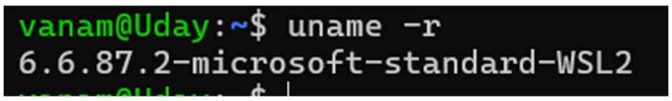
uname — Get kernel and system information

Option	Meaning
-a	Display all available system information
-r	Kernel release
-m	Machine hardware architecture
-s	Kernel name
-n	Network hostname





R



M



ip — Inspect network interfaces and addresses

Option or subcommand	Meaning
addr	Show IP addresses and interfaces
-br	Brief, compact output
route	Display routing information

link **Display network-interface information**

```
vanam@Uday:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet 10.255.255.254/32 brd 10.255.255.254 scope global lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:15:5d:09:0e:f4 brd ff:ff:ff:ff:ff:ff
    inet 172.19.186.218/20 brd 172.19.191.255 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::215:5dff:fe09:ef4/64 scope link
       valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether 72:3e:08:3a:1a:ee brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
       valid_lft forever preferred_lft forever
```

Addr

```
vanam@Uday:~$ ip -br addr
lo               UNKNOWN        127.0.0.1/8 10.255.255.254/32 ::1/128
eth0             UP             172.19.186.218/20 fe80::215:5dff:fe09:ef4/64
docker0          DOWN           172.17.0.1/16
```

Curl

```
vanam@Uday:~$ curl https://example.com
<!doctype html><html lang=en><head><meta charset=utf-8><link rel=icon href=data:,><meta name=viewport content="width=dev
ice-width,initial-scale=1"><title>Example Domain</title><style>html{color-scheme:light dark;background:light-dark(#eee,#
222)}body{font:16px/1.6 system-ui,sans-serif;max-width:26em;margin:auto;padding:25vh 2em 2em;text-align:center}</style><
/head><body><p>This domain is for use in documentation examples without needing permission. This is not a service; avoid
 relying on it for testing and monitoring purposes.</p><script src=/s.js></script></body></html>
```

jobs — Inspect background jobs

```
vanam@Uday:~$ sleep 200 &
jobs
jobs -l
[1] 840
[1]+  Running                 sleep 200 &
[1]+   840 Running                sleep 200 &
```